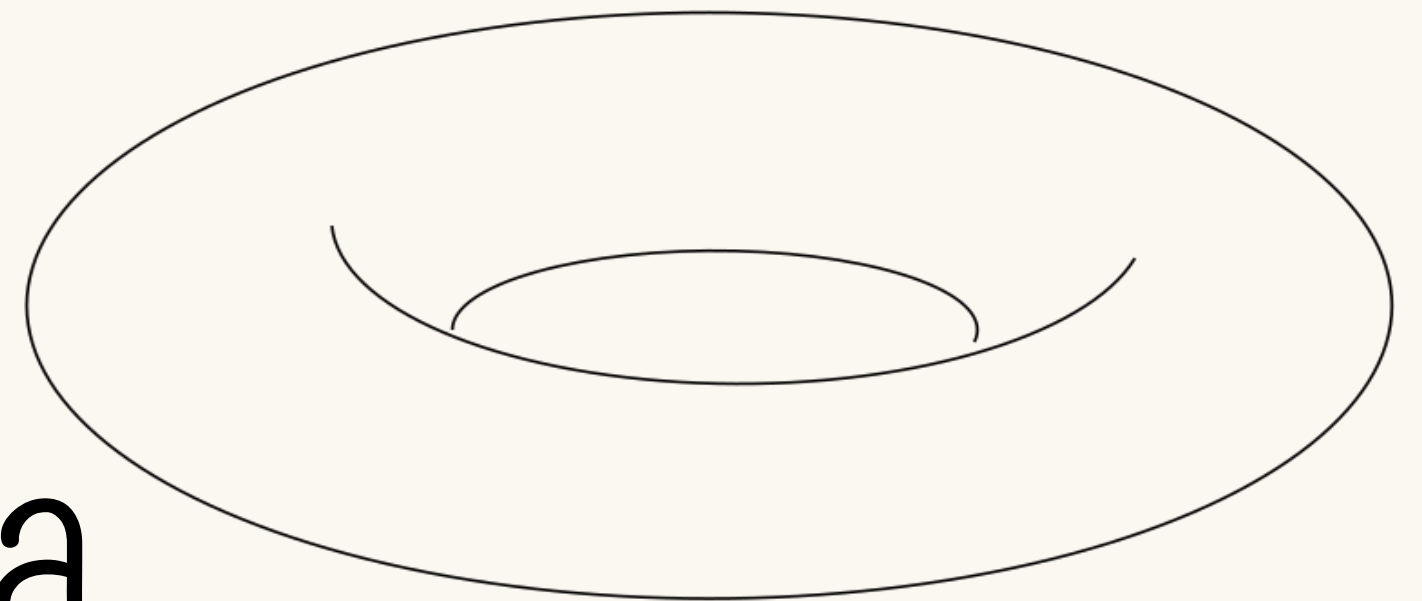


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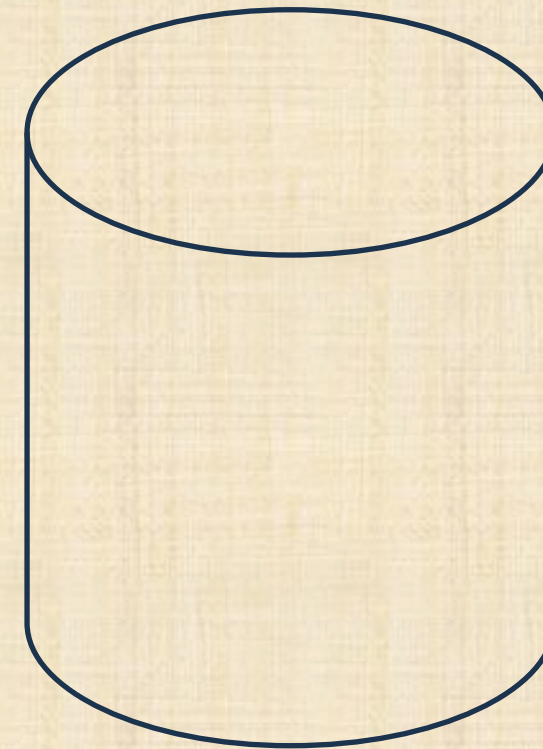
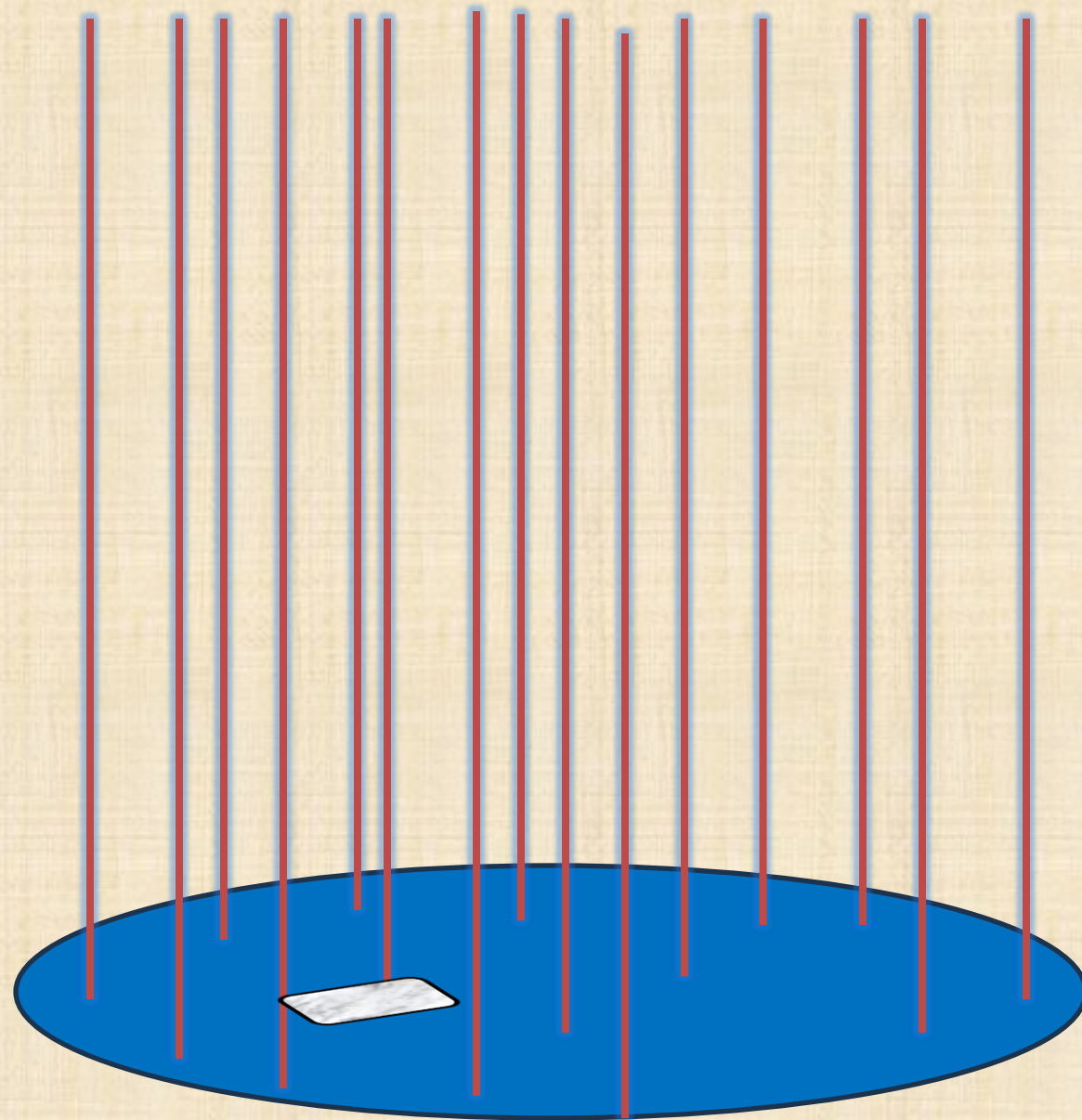
“FIBER” BUNDLES

in

topological data
analysis *



What is a fiber bundle?



Definitions

Fiber:

Suppose $\pi : E \rightarrow B$ is a map (between topological spaces). Any subset of the form $E_p := \pi^{-1}(p) \subseteq E$ for each $p \in B$ is called a *fiber* of E .

If π is surjective, then each fiber is non-empty and π partitions the domain E , so that $E = \sqcup_{p \in B} E_p$.

Fiber bundle:

Given two topological spaces E and B , a fiber bundle over B with fiber F is a surjective map $\pi : E \rightarrow B$ such that:

1. for each $p \in B$, $\pi^{-1}(p) = F$,
2. there exists an open subset $U \subseteq B$ containing p and a homeomorphism $\phi : \pi^{-1}(U) \rightarrow U \times F$ (called a *local trivialization* of E over U), such that $\pi_U \circ \phi = \pi$ (where $\pi_U : U \times F \rightarrow U$ is the projection onto U).

We refer to E as the “*total space*”, F as the “*fiber*”, and B as the “*base space*”.

Definitions

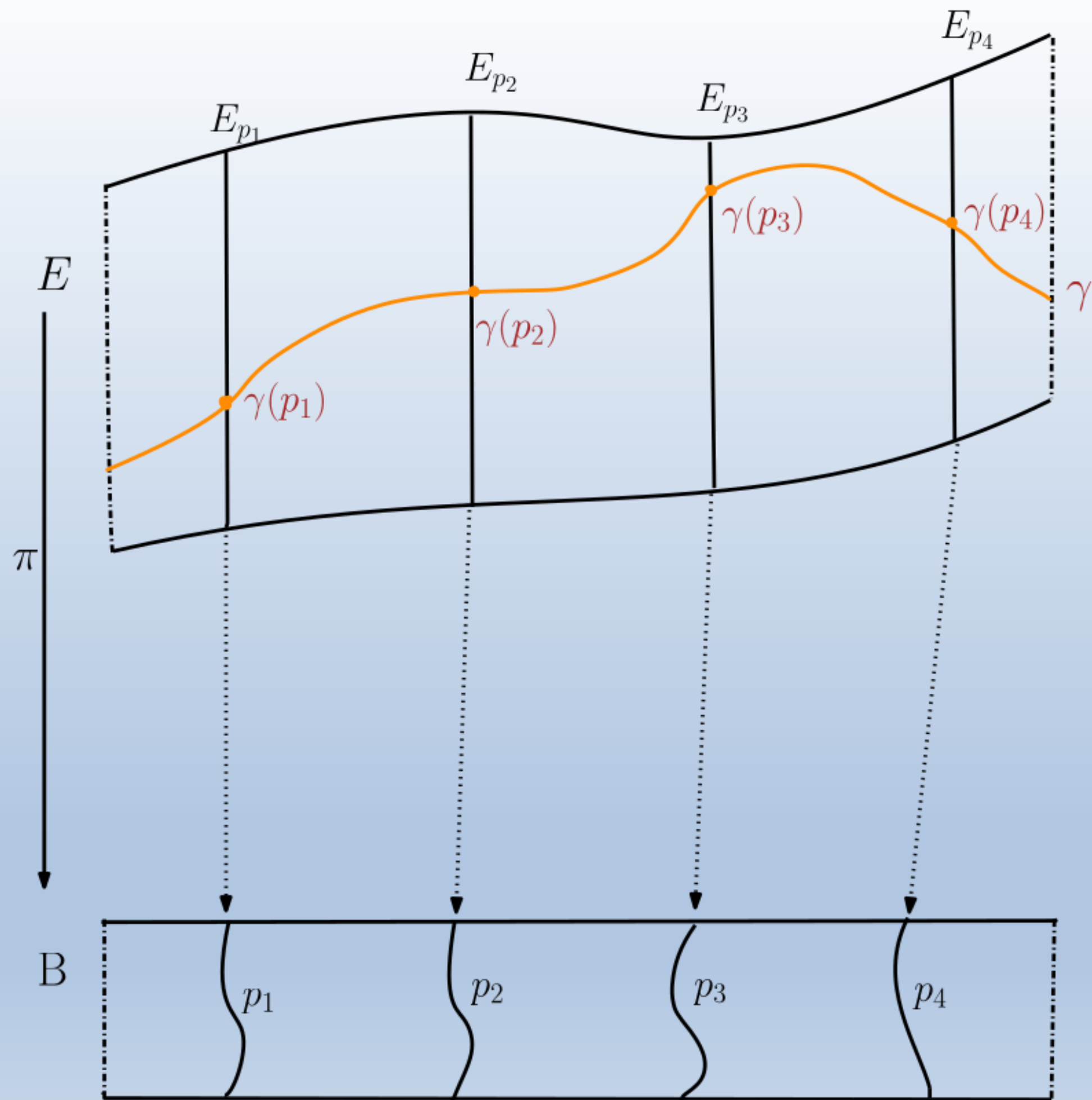
Sections:

A *section* of the fiber bundle $\pi : E \rightarrow B$ is a continuous map $\gamma : B \rightarrow E$ such that the projection of γ is the identity. That is, $\pi \circ \gamma(p) = p$ for each p in B .

The section γ is said to be *local* if it is only defined on an open subset U of B .

Transition functions:

Let $\phi_1 : \pi^{-1}(U_1) \rightarrow U \times F$ and $\phi_2 : \pi^{-1}(U_2) \rightarrow U \times F$ be two local trivializations of a fiber bundle $\pi : E \rightarrow B$ with the same fiber F . The transition function from ϕ_1 to ϕ_2 is $g_{21} : U_1 \cap U_2 \rightarrow \text{Aut}(F)$ defined by $(\phi_2 \circ \phi_1^{-1})(p, x) = (p, g_{21}(p)(x))$ for $p \in U_1 \cap U_2$ and $x \in F$.



Examples of fiber bundles

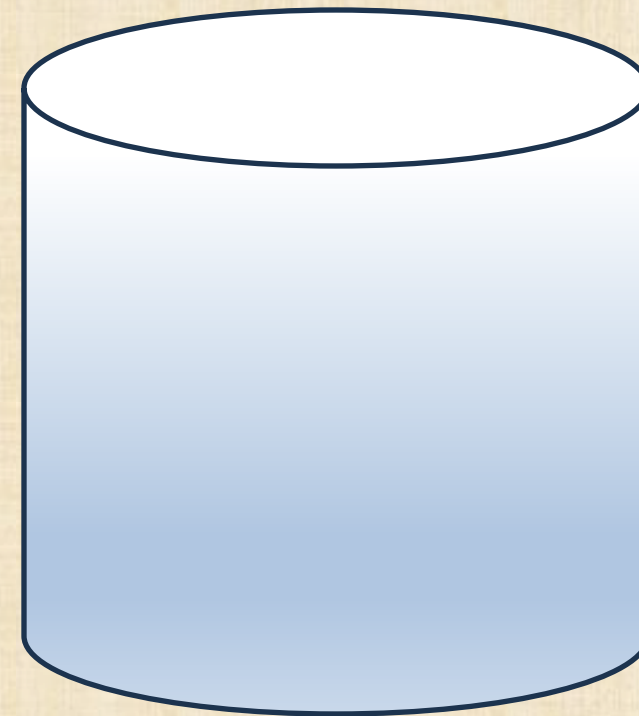
Mobius band

$$B = S^1$$
$$F = [0, 1]$$



(Hollow) cylinder

$$B = S^1$$
$$F = [0, 1]$$



Examples of fiber bundles

Klein bottle

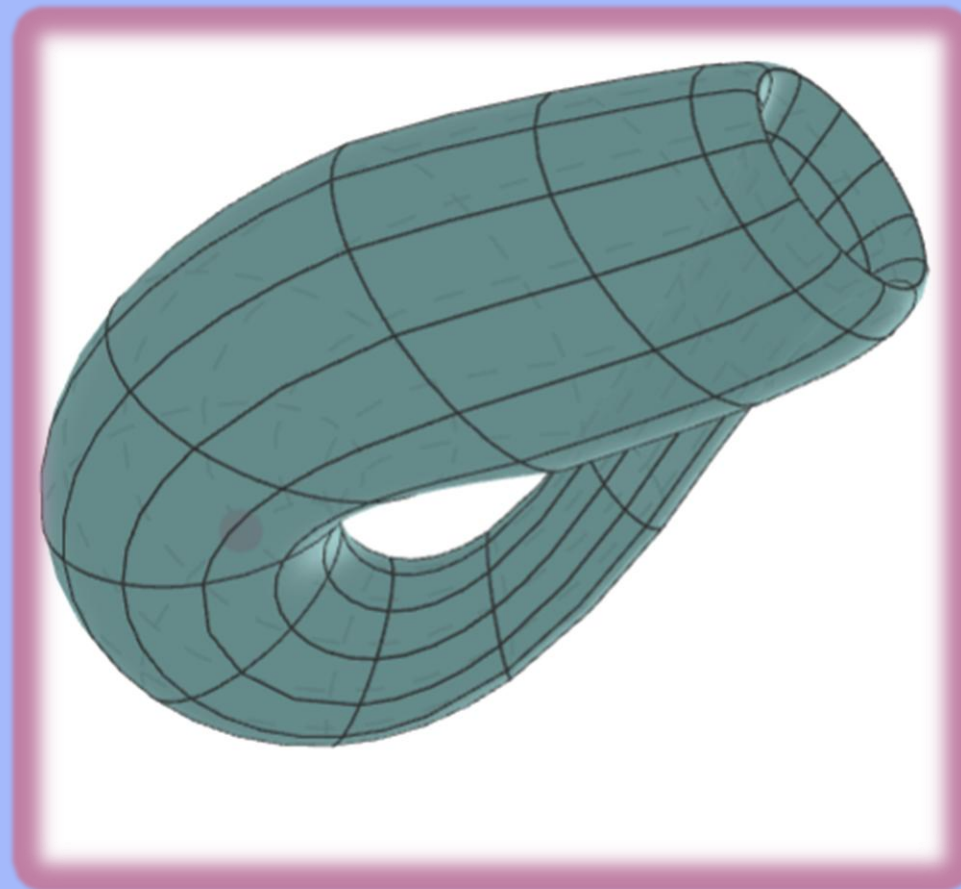
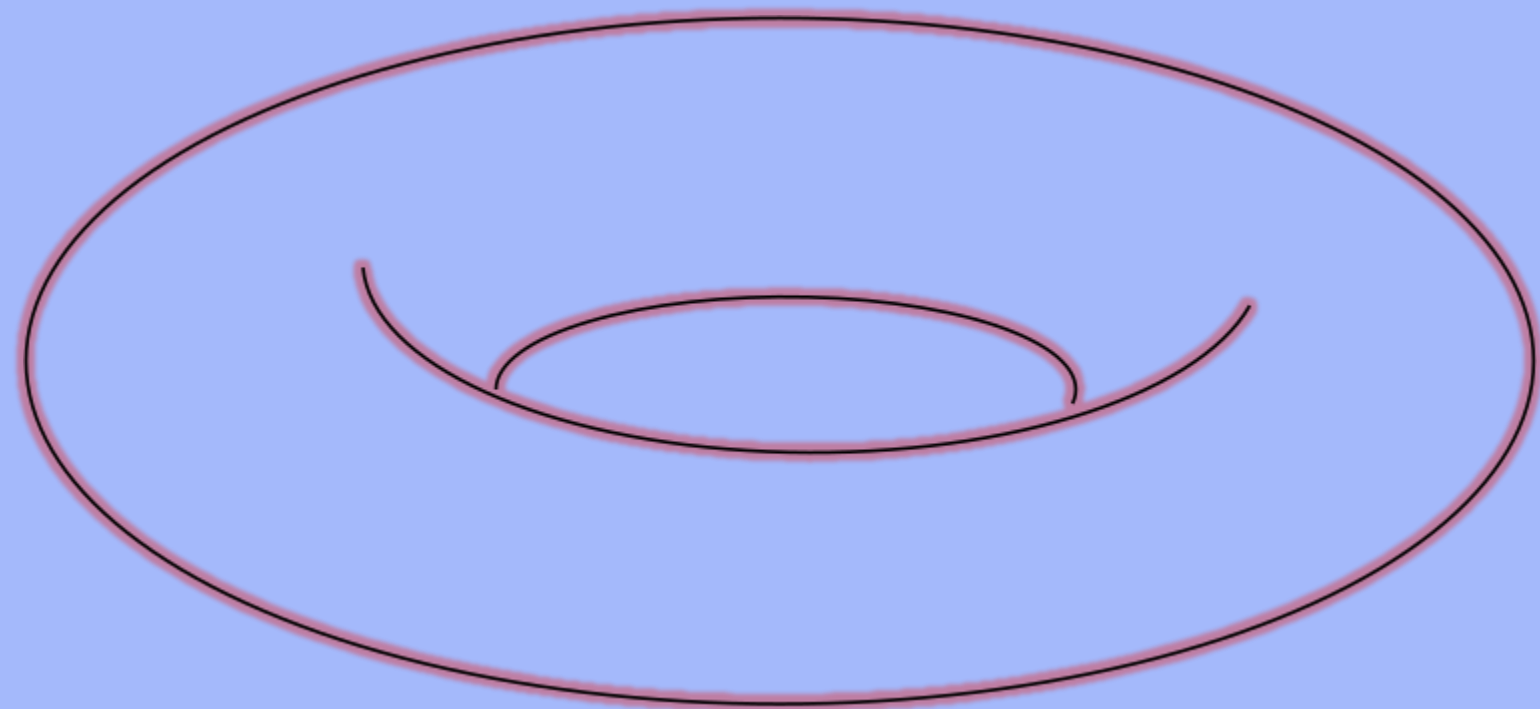


Image credit: "Klein bottle", a Geogebra applet
by Juan Carlos Ponce Campuzano,
www.geogebra.org/u/jcponce/

$$\begin{aligned} B &= S^1 \\ F &= S^1 \end{aligned}$$

Torus



$$\begin{aligned} B &= S^1 \\ F &= S^1 \end{aligned}$$

Persistence diagram (PD) bundles'

Let $\mathcal{PD}$ denote the space of persistence diagrams. A *persistence diagram bundle* consists of a topological space, B , as the base space, and a function $f : B \rightarrow \mathcal{PD}$ which is continuous with respect to the Bottleneck distance. The total space, E , is given by

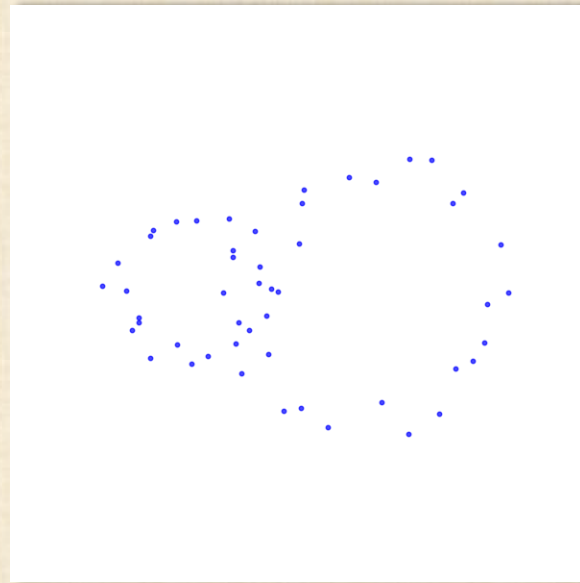
$$E := \{(p, z) : p \in B, z \in f(p)\},$$

and has the subspace topology induced by the inclusion $E \subseteq B \times \mathcal{PD}$.

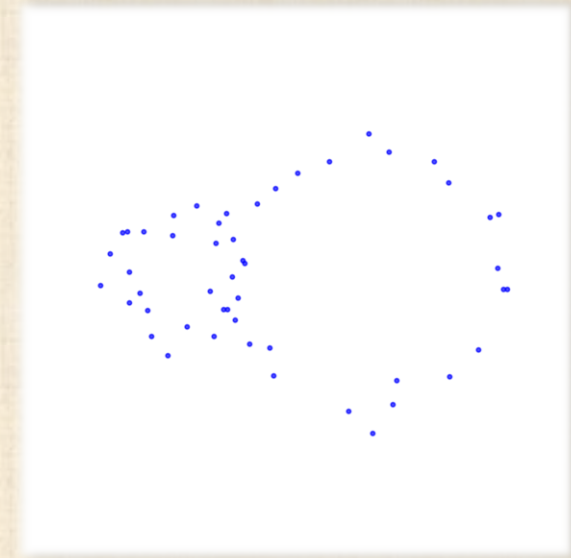
Sections A *section* within a persistence diagram bundle is a continuous function $\gamma : B \rightarrow E$ such that $\gamma(p) \in f(p)$ for all $p \in B$, and the set $\{p \in B : \gamma(p) \neq \Delta\}$ is connected.

Examples of PD bundles

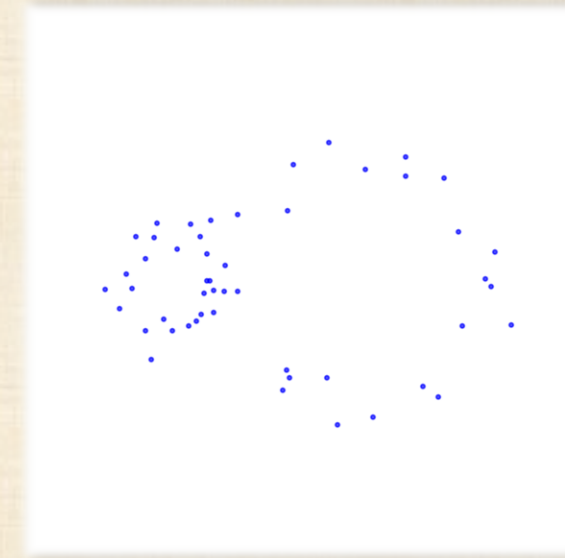
Setup: Persistent homology of evolving object



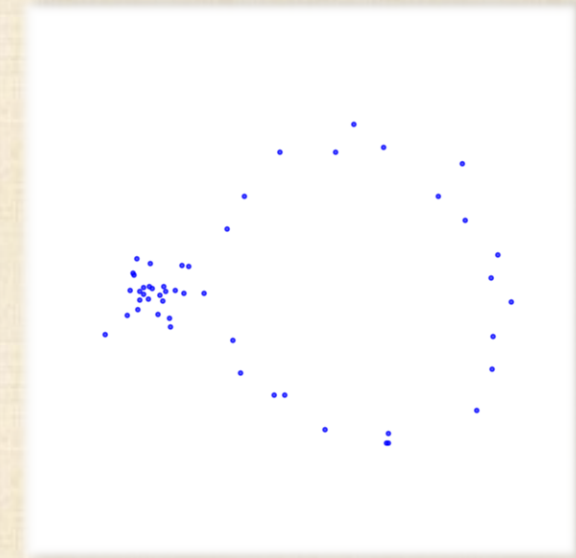
t_0



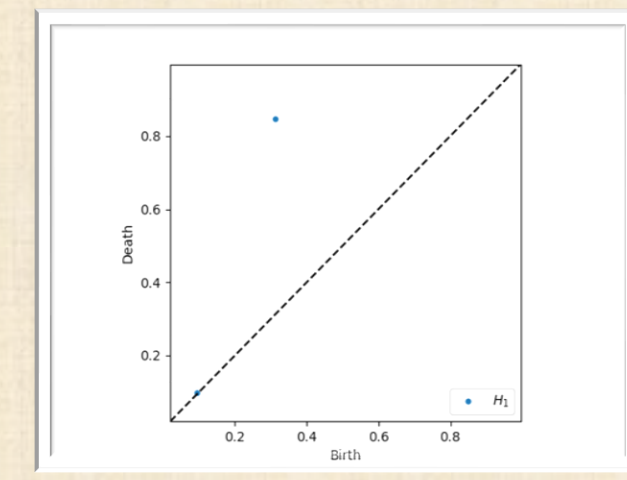
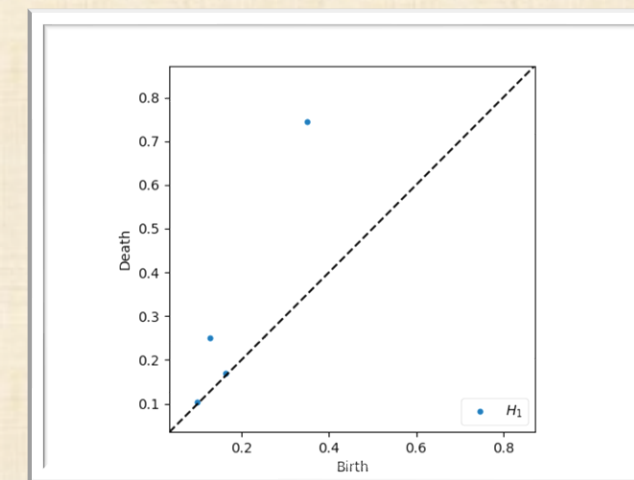
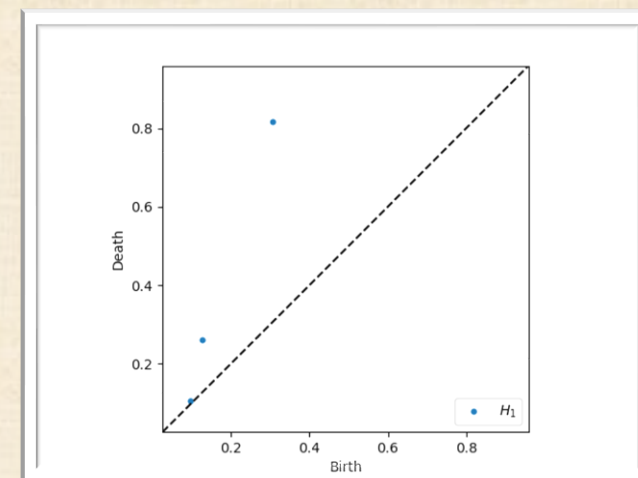
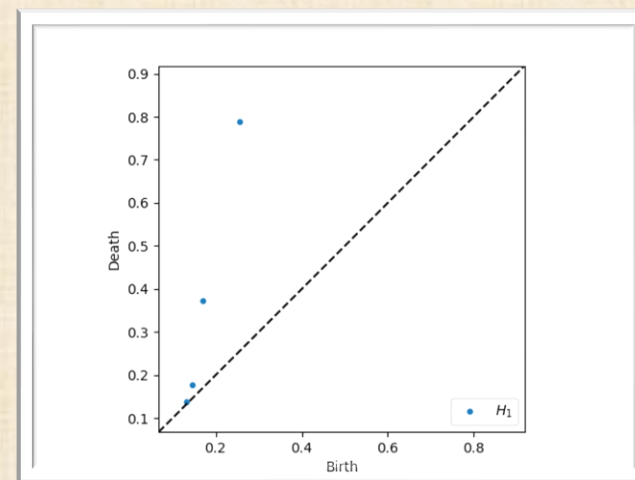
t_1



t_2

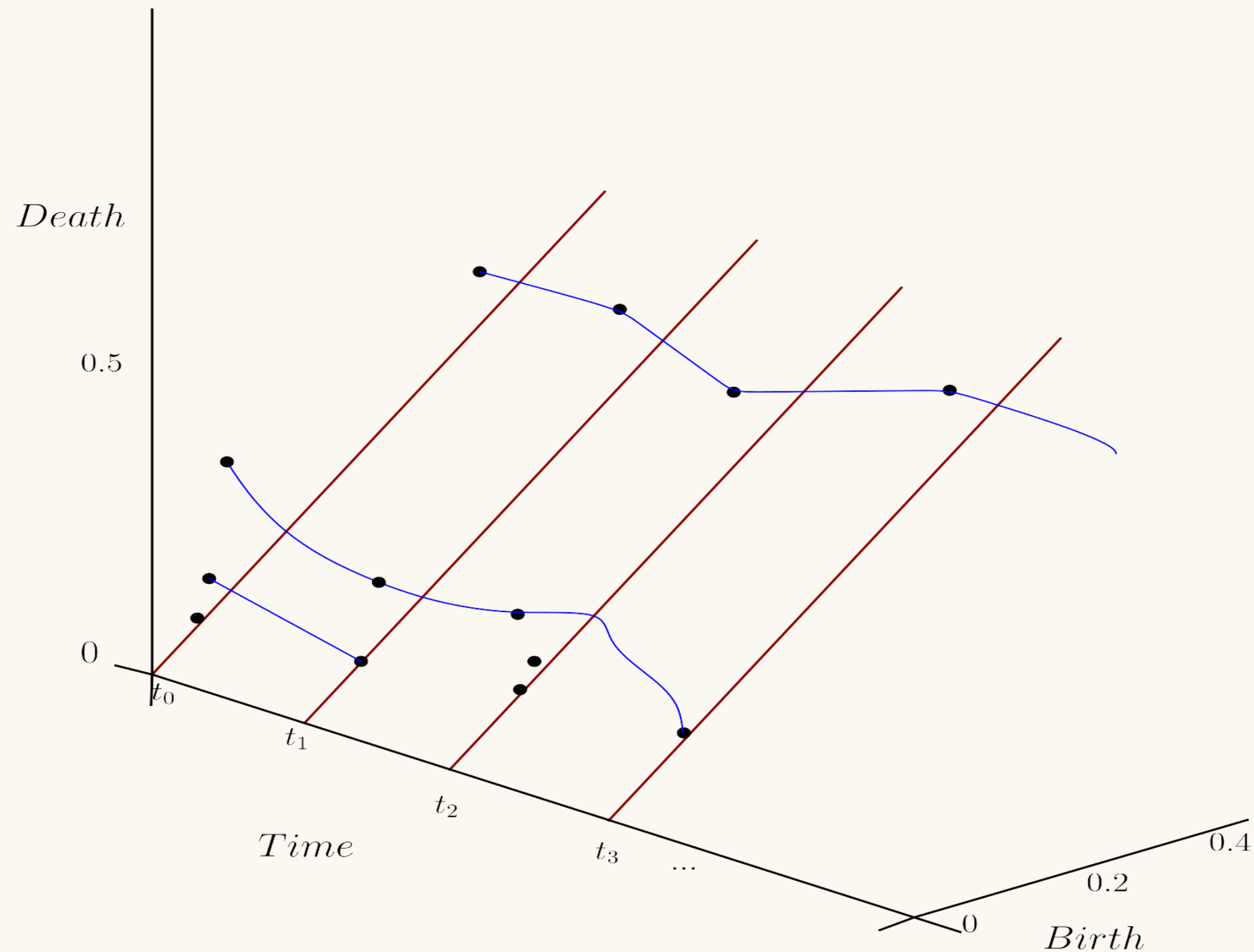


t_3



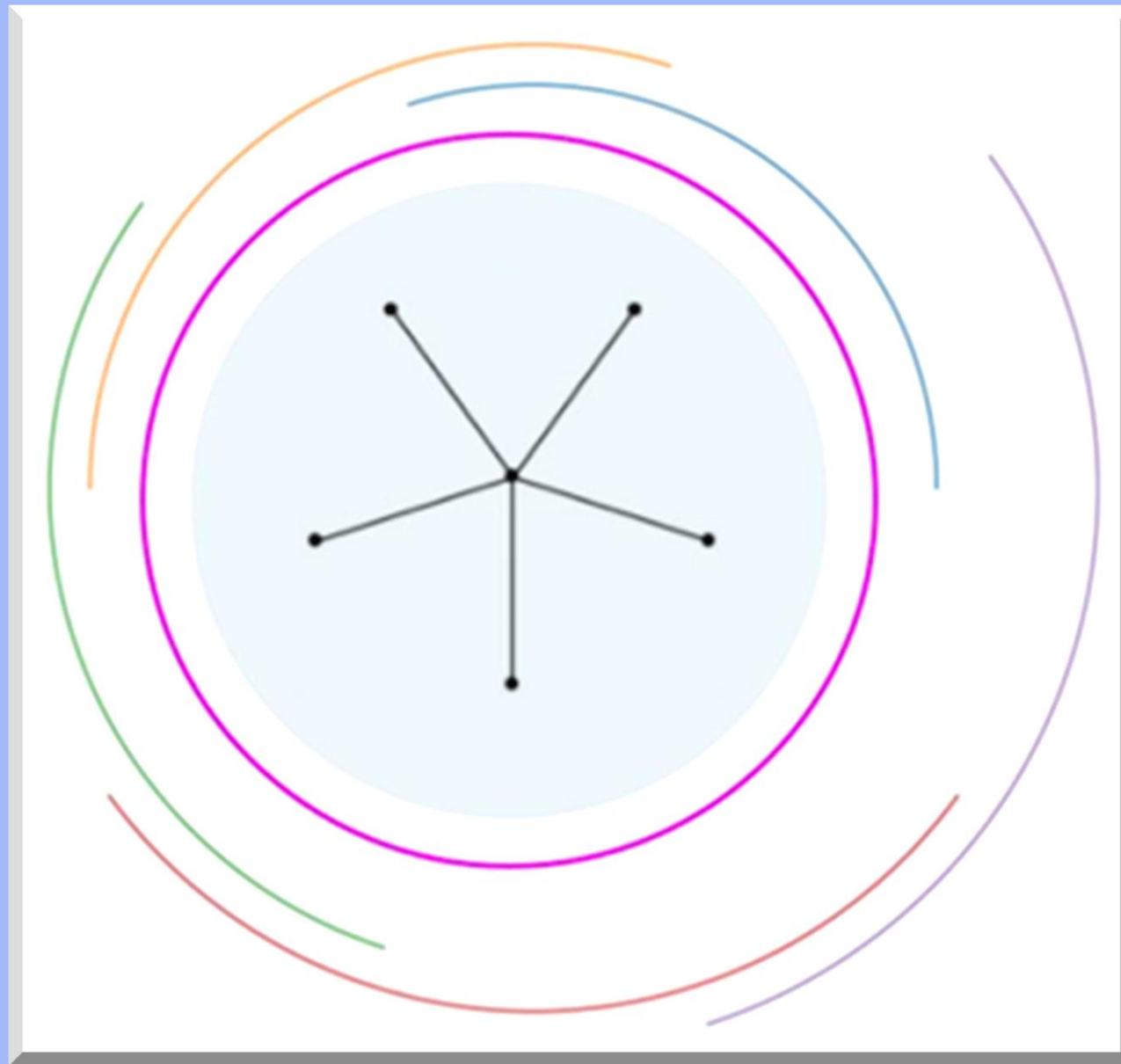
Examples of PD bundles

Topological persistence vineyard²



Examples of PD bundles

Persistent homology transform³



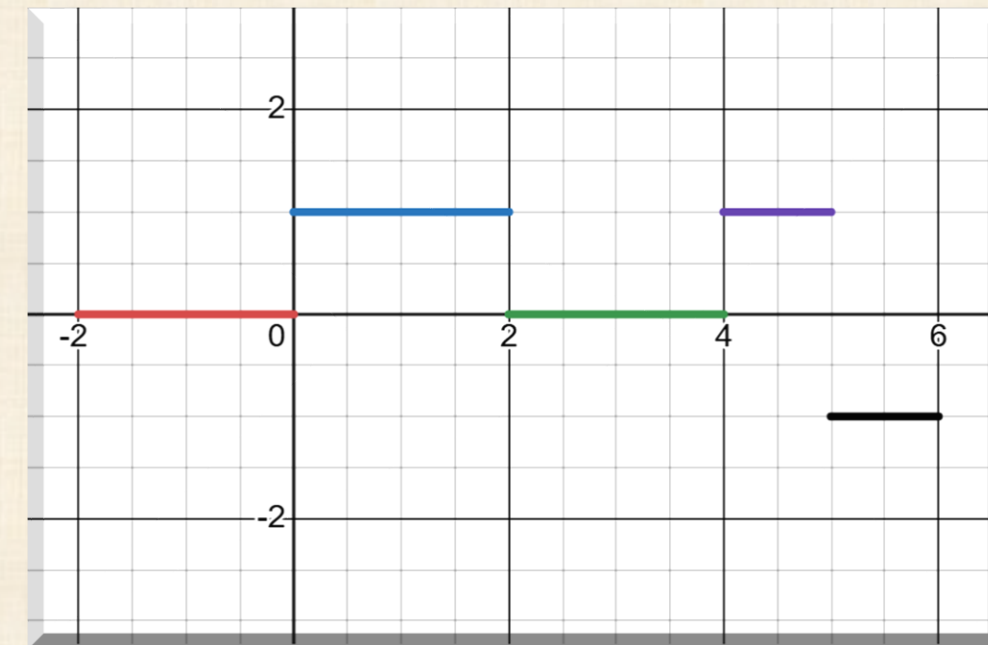
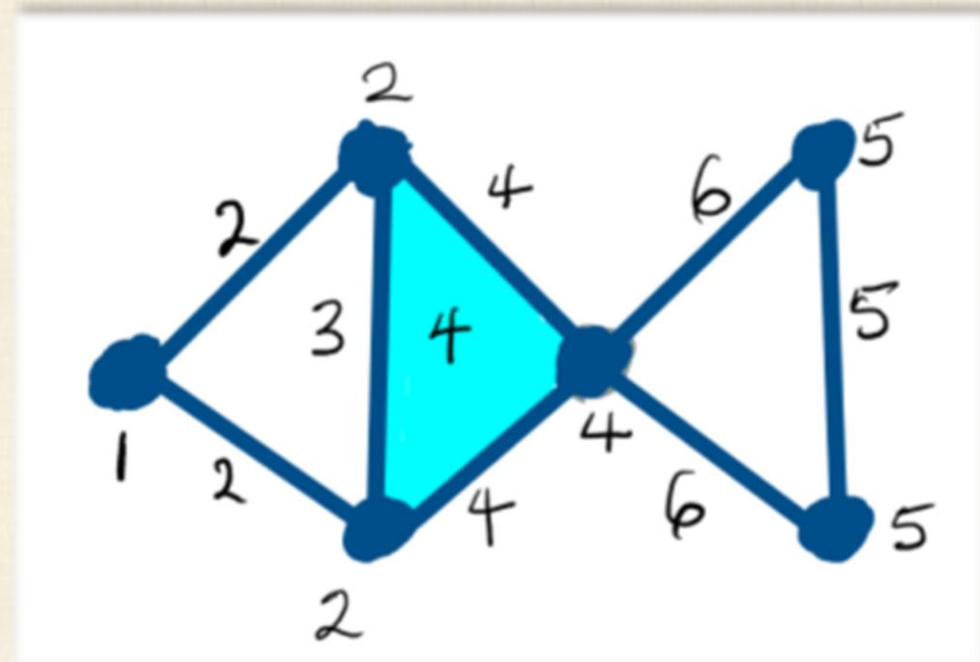
PHT₀ of star-shaped object
Image credit:⁴ figure 2(b) of [1]

$B = S^{d-1}$
 F : persistence diagram

3. Turner et al, 2014 [6].
4. Arya et al, 2024 [1].

Example of EC bundles

Euler characteristic transform (ECT)⁴



Euler characteristic curve (ECC)

Another directional transform, like the PHT

$B = S^{d-1}$
 $F : \text{Euler characteristic curve}$

Sections capture the evolution of an Euler characteristic value along directions on the sphere

The ECT (as well as the PHT) is injective on the space of constructible sets in $\mathbb{R}^d$ [6].

References

- [1] Shreya Arya, Barbara Giunti, Abigail Hickok, Lida Kanari, Sarah McGuire, and Katharine Turner. Decomposing the persistent homology transform of star-shaped objects, 2024, arXiv:2408.14995.
- [2] David Cohen-Steiner, Herbert Edelsbrunner, and Dmitriy Morozov. Vines and vineyards by updating persistence in linear time. In *Proceedings of the Twenty-Second Annual Symposium on Computational Geometry, SCG '06*, page 119–126, New York, NY, USA, 2006. Association for Computing Machinery.
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- [4] Abigail Hickok. Computing persistence diagram bundles, 2023, arXiv:2210.06424.
- [5] Elizabeth Munch. An invitation to the euler characteristic transform. *The American Mathematical Monthly*, 132(1):15–25, 2025, <https://doi.org/10.1080/00029890.2024.2409616>.
- [6] Katharine Turner, Sayan Mukherjee, and Doug Boyer. Persistent homology transform for modeling shapes and surfaces. *Information and Inference*, 3:310–344, 01 2014.

